Do ghosts prefer schools, hotels or cemeteries?

Patterns in a collection of US haunting reports

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Introduction

This project asks which locations—schools, hotels or cemeteries—appear most often in a public collection of US haunting reports, and how the phenomena mentioned in their descriptions differ. It examines patterns in stories, not whether ghosts exist. Without totals of eligible buildings, report shares cannot measure haunting probabilities.

Data sample

TidyTuesday's *Haunted Places in the United States* dataset contains 10,992 records compiled by Tim Renner from the Shadowlands Index [1], with location names, descriptions, cities, states and coordinates. Removing two exact duplicates leaves 10,990 records. Missing text values are replaced by empty strings; coordinates are not used.

Case-insensitive whole-word searches classify location names. Schools include universities, colleges and academies; hotels include motels, inns and resorts; cemeteries include graveyards and burial grounds. Excluding six ambiguous matches and 7,475 unmatched records leaves 3,509 reports (31.9% of the cleaned collection).

Data analysis method

Python uses pandas for processing and Matplotlib for plotting. Category shares divide counts by 3,509; phenomenon percentages divide keyword matches by reports *within each category*. Descriptions can match multiple groups. Moving-object matches require an object-related word followed within 50 characters, without crossing sentence-ending punctuation, by a movement-related word.

For k matches in n reports, $\hat{p} = k/n$ and $\text{SE} = \sqrt{\hat{p}(1-\hat{p})/n}$. Individual 95% Wilson intervals use $z = 1.959964$. These model-based intervals assume independent reports and exclude reporting bias and classification errors. Binary sample SD is also exported, separately from SE. A sample of up to ten reports per category is exported for manual review; classification accuracy has not been validated.

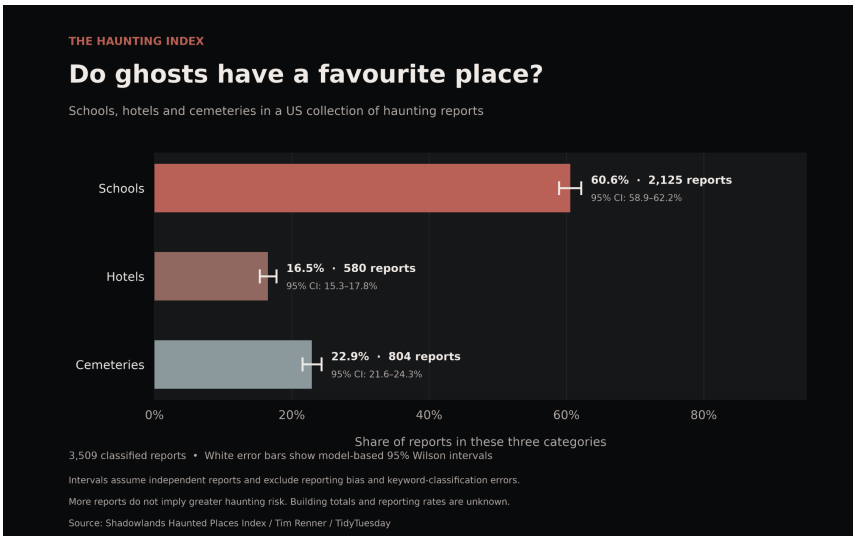


Figure 1: Shares of the 3,509 selected reports. White bars show individual 95% Wilson intervals. Schools are the largest category; the graph does not measure haunting risk.

Results and discussion

Schools account for 60.6% of the selected reports (95% Wilson interval: 58.9–62.2%), compared with 22.9% for cemeteries (21.6–24.3%) and 16.5% for hotels (15.3–17.8%). Figure 1 shows the counts as shares of the same denominator. There are approximately 3.7 times as many school reports as hotel reports and 2.6 times as many as cemetery reports. These ratios describe representation in this collection, not relative haunting risk.

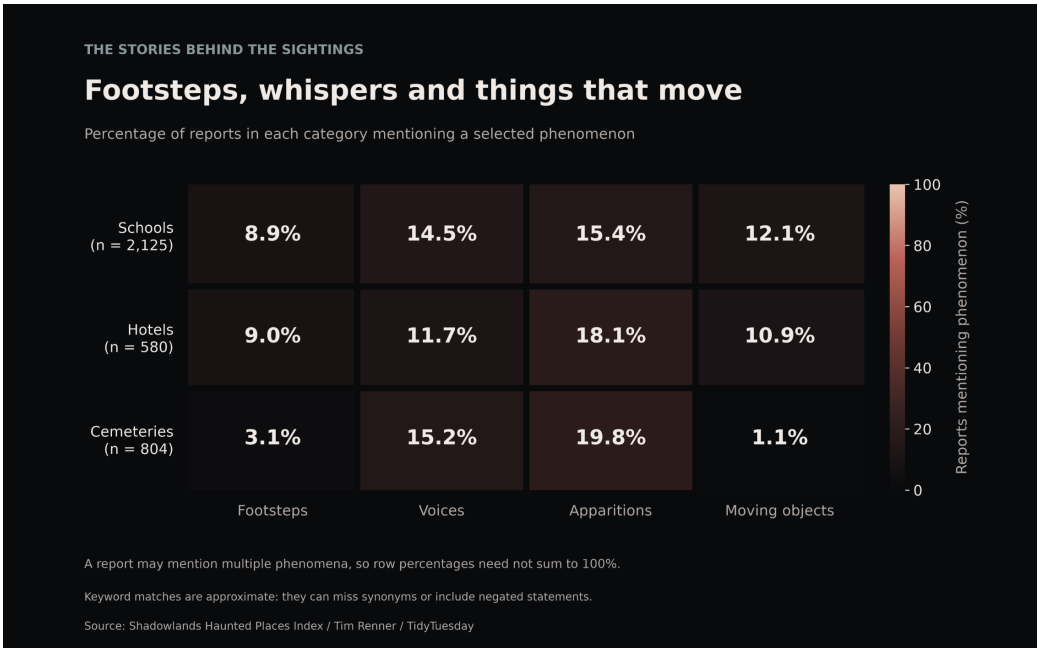


Figure 2: Keyword matches within each location category. Reports may mention several phenomena or none, so row percentages need not sum to 100%. Cells show percentages, not verified sightings.

Description keywords also differ between settings (Figure 2). Apparition-related keywords are the most common of the four groups in each category. Moving-object keywords appear in 257 of 2,125 school reports (12.1%; Wilson interval 10.8–13.5%) and 9 of 804 cemetery reports (1.1%; 0.6–2.1%), a descriptive difference of about 11 percentage points. Footsteps are mentioned in about 9% of school and hotel reports but 3.1% of cemetery reports. The analysis does not test a causal explanation for these contrasts.

Voluntary reporting, unknown building totals and possible shared stories limit generalisation. Keyword searches can miss synonyms, include negations and misclassify names. Individual confidence intervals are not adjusted for multiple comparisons.

In conclusion, schools dominate the three-category subset, and the descriptions differ in keyword content. These are reproducible patterns under the stated selection rules, not evidence that ghosts prefer particular locations.

References

[1] TidyTuesday (2023). *Haunted Places in the United States*, 10 October dataset. Compiled by Tim Renner from the Shadowlands Index.
https://github.com/rfordatascience/tidytuesday/tree/main/data/2023/2023-10-10